

EC602 — Computer Network

Semester VI • 3L:0T:0P • 3 Credits

Module 3 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Network Layer

Network Layer (Layer 3): Responsible for *source-to-destination* delivery of packets across multiple networks. Handles logical addressing (IP), routing, and internetworking.

1.1 Internetworking Devices

Device	OSI Layer	Function
Repeater	Layer 1	Regenerates and amplifies signals; extends cable length. No filtering.
Hub	Layer 1	Multi-port repeater; broadcasts to all ports. No intelligence.
Bridge	Layer 2	Connects two LAN segments; filters frames using MAC address table.
Switch	Layer 2	Multi-port bridge; creates dedicated paths between ports; reduces collisions.
Router	Layer 3	Routes packets between different networks using IP addresses and routing tables.
Gateway	Layer 4–7	Protocol converter; connects networks using different protocols (e.g. LAN to mainframe).

1.2 IP Addressing

IP Address: A 32-bit (IPv4) logical address assigned to each device on a network. Written in dotted-decimal notation (e.g. 192.168.1.1). Consists of **Network ID** and **Host ID**.

IPv4 Address Classes:

Class	Range (1st octet)	Default Mask	Use
A	1 – 126	255.0.0.0	Large networks; 126 networks, ~16M hosts each
B	128 – 191	255.255.0.0	Medium networks; 16384 networks, 65534 hosts each
C	192 – 223	255.255.255.0	Small networks; ~2M networks, 254 hosts each
D	224 – 239	—	Multicast
E	240 – 255	—	Reserved / experimental

1.3 Subnetting

Subnetting

Concept: Dividing a large network into smaller sub-networks (subnets) by borrowing bits from the Host ID portion.

Subnet Mask: A 32-bit number with 1s for network+subnet bits and 0s for host bits.

Key formulas:

- Number of subnets = 2^s (where s = borrowed bits)
- Hosts per subnet = $2^h - 2$ (where h = remaining host bits; subtract 2 for network and broadcast addresses)
- Network address: AND of IP address with subnet mask
- Broadcast address: set all host bits to 1

Subnetting Example

Given: Class C network 192.168.1.0, subnet mask 255.255.255.192 (/26).

Borrowed bits $s = 2$, host bits $h = 6$.

Subnets = $2^2 = 4$. Hosts/subnet = $2^6 - 2 = 62$.

Subnets: 192.168.1.0/26, 192.168.1.64/26, 192.168.1.128/26, 192.168.1.192/26.

1.4 Routing

Routing: The process of selecting the best path for packets to travel from source to destination across an internetwork. Performed by routers using **routing tables**.

Static vs. Dynamic Routing

Parameter	Static Routing	Dynamic Routing
Configuration	Manually by admin	Automatically by routing protocols
Adaptability	Cannot adapt to changes	Adapts to topology changes
Overhead	No routing protocol overhead	Routing protocol traffic overhead
Scalability	Poor for large networks	Good; scales well
Use case	Small, stable networks	Large, dynamic networks

1.5 Unicast Routing Protocols

RIP (Routing Information Protocol)

RIP — Distance Vector Protocol

Algorithm: Bellman-Ford. Each router maintains a table of (destination, next-hop, distance) and shares it with **direct neighbours** every 30 seconds.

Metric: Hop count. **Maximum hops:** 15 (16 = infinity/unreachable).

Versions: RIPv1 (classful, broadcast), RIPv2 (classless, multicast 224.0.0.9).

Problems: Slow convergence, count-to-infinity problem, not suitable for large networks.

OSPF (Open Shortest Path First)

OSPF — Link State Protocol

Algorithm: Dijkstra's Shortest Path First. Each router floods **Link State Advertisements (LSAs)** to all routers; every router builds a complete topology map.

Metric: Cost (based on bandwidth). **No hop limit.**

Features: Fast convergence, hierarchical (areas), supports VLSM/CIDR, authentication.

Hello packets: Sent every 10 seconds to discover and maintain neighbours.

BGP (Border Gateway Protocol)

BGP — Path Vector Protocol

Use: Inter-domain routing between Autonomous Systems (AS) on the Internet. The Internet's core routing protocol.

Algorithm: Path vector; each route advertisement includes the full AS path.

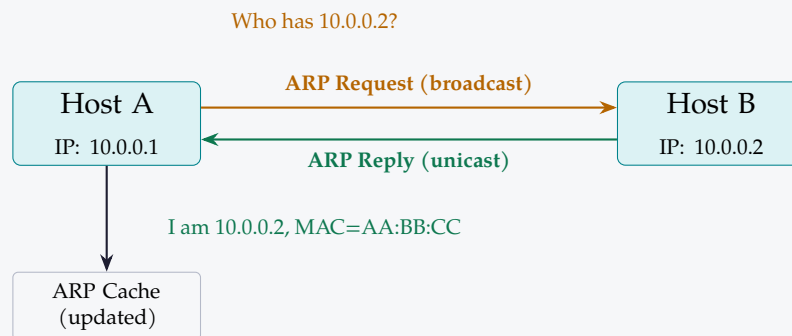
Metric: Policy-based (AS path length, local preference, MED, etc.).

Port: TCP port 179. **Types:** iBGP (within AS), eBGP (between AS).

1.6 Other Network Layer Protocols

Protocol	Function
ARP	Address Resolution Protocol. Maps IP address → MAC address. Broadcasts ARP request; target replies with its MAC.
RARP	Reverse ARP. Maps MAC address → IP address. Used by diskless workstations (replaced by DHCP).
ICMP	Internet Control Message Protocol. Reports errors and diagnostic info (e.g. ping uses ICMP Echo Request/Reply).
IP (IPv4)	Connectionless, unreliable packet delivery. 32-bit addressing, fragmentation, TTL field.
IPv6	128-bit addressing (340 undecillion addresses), simplified header, no fragmentation by routers, built-in IPSec.

ARP — Address Resolution Process



IPv4 vs. IPv6 Key Differences:

Feature	IPv4	IPv6
Address size	32 bits (4 bytes)	128 bits (16 bytes)
Notation	Dotted decimal (x.x.x.x)	Colon-hex (x:x:x:x:x:x:x:x)
Header size	20–60 bytes (variable)	40 bytes (fixed)
Fragmentation	By routers and hosts	By source host only
Checksum	Present in header	Removed (handled by upper layers)
Security	Optional (IPSec)	Mandatory (IPSec built-in)

2. Transport Layer

Transport Layer (Layer 4): Provides *process-to-process* (end-to-end) delivery of data. Handles segmentation, reassembly, flow control, error control, and multiplexing using **port numbers**.

2.1 UDP (User Datagram Protocol)

UDP — Connectionless, Unreliable

Features: No connection setup, no acknowledgement, no flow/error control. Minimal overhead.

Header: 8 bytes — Source Port (16b) | Destination Port (16b) | Length (16b) | Checksum (16b).

Use cases: DNS, DHCP, SNMP, VoIP, video streaming, online gaming.

Advantage: Very fast, low latency. **Disadvantage:** No reliability guarantee.

2.2 TCP (Transmission Control Protocol)

TCP — Connection-Oriented, Reliable

Features: Full-duplex, byte-stream, reliable delivery, flow control (sliding window), congestion control, ordered delivery.

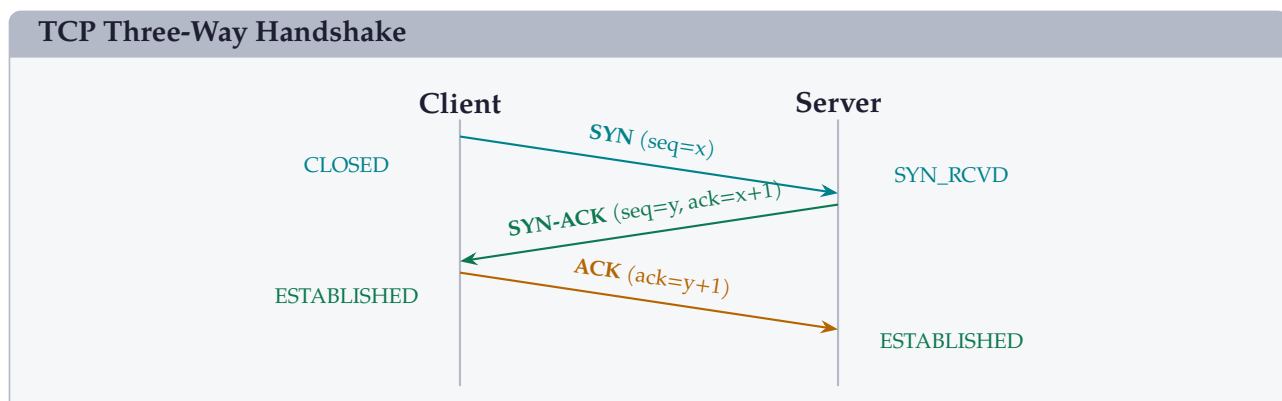
Header: 20–60 bytes. Key fields: Source/Dest Port, Sequence No., ACK No., Flags (SYN, ACK, FIN, RST), Window Size, Checksum.

Use cases: HTTP, FTP, SMTP, SSH, Telnet.

TCP Three-Way Handshake (Connection Establishment):

1. **SYN:** Client sends SYN segment with initial sequence number x .
2. **SYN-ACK:** Server responds with SYN (y) + ACK ($x + 1$).
3. **ACK:** Client sends ACK ($y + 1$). Connection established.

TCP Connection Termination (Four-Way): FIN → ACK → FIN → ACK.



2.3 Congestion Control

Congestion: A state where the network cannot handle the offered load; routers drop packets, causing retransmissions and further congestion. Congestion control prevents/reduces this.

Open Loop Congestion Control

Prevents congestion **before it occurs** through good design policies:

- **Retransmission policy:** Retransmit only when necessary (use timers wisely).
- **Window policy:** Use selective repeat (wastes less bandwidth than GBN).
- **Discarding policy:** Routers discard less sensitive packets first.
- **Acknowledgement policy:** Receiver delays ACK to reduce ACK traffic.
- **Admission policy:** Routers reject new connections if network is near capacity.

Closed Loop Congestion Control

Reacts to congestion **after it occurs**:

- **Backpressure:** Congested node sends a signal to upstream node to slow down. Works hop-by-hop.
- **Choke packet:** Congested router sends a choke packet directly to the source to reduce its transmission rate.
- **Implicit signalling:** Source detects congestion from dropped packets or increased delay (TCP's approach).
- **Explicit signalling:** Router sets a bit in the packet header to warn source/destination (ECN — Explicit Congestion Notification).

2.4 Quality of Service (QoS)

QoS: The ability of a network to provide different levels of service to different traffic types. Key metrics: **bandwidth**, **delay**, **jitter** (delay variation), **packet loss**.

Leaky Bucket Algorithm

Leaky Bucket Algorithm

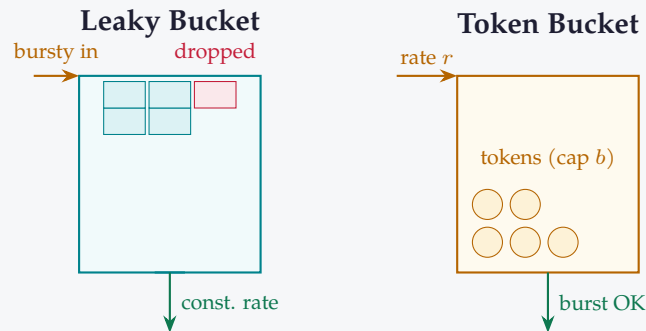
Concept: A bucket with a small hole at the bottom. Packets enter the bucket (queue) at any rate; they leave at a **constant rate** (the leak rate). Smooths bursty traffic into a steady stream.

Operation:

- If bucket is not full: incoming packet is added to the queue.

- If bucket is full: packet is **discarded** (overflow).
 - Packets leave the bucket at a fixed rate r (tokens/second).
- Advantage:** Enforces constant output rate. **Disadvantage:** Cannot handle bursts; excess packets dropped.

Leaky Bucket vs Token Bucket



Token Bucket Algorithm

Token Bucket Algorithm

Concept: Tokens are generated at a constant rate r and stored in a bucket (capacity b). A packet can only be sent if a token is available (one token consumed per packet).

Operation:

- Tokens accumulate up to bucket capacity b .
- To send a packet: remove one token from bucket.
- If no token available: packet must wait (or be discarded).

Maximum burst size: b packets (when bucket is full).

Advantage: Allows controlled bursts up to b . More flexible than leaky bucket.

Feature	Leaky Bucket	Token Bucket
Output rate	Constant (fixed)	Variable (up to burst limit)
Burst handling	Excess packets dropped	Bursts allowed up to b tokens
Flexibility	Less flexible	More flexible
Use case	Traffic shaping	Traffic policing with burst allowance

Quick Revision — Module 3

Network Layer & Transport Layer | EC602

Key Facts at a Glance

Class A range	1–126; mask 255.0.0.0
Class B range	128–191; mask 255.255.0.0
Class C range	192–223; mask 255.255.255.0
Subnets	2^s (s = borrowed bits)
Hosts per subnet	$2^h - 2$ (h = host bits)
RIP metric / max hops	Hop count / 15
OSPF metric	Cost (bandwidth-based)
BGP use	Inter-AS routing; TCP port 179
ARP function	IP → MAC address resolution
ICMP use	Error reporting; ping (Echo Request/Reply)
IPv6 address size	128 bits
TCP handshake	SYN → SYN-ACK → ACK
UDP header size	8 bytes
TCP header size	20–60 bytes
Leaky bucket output	Constant rate
Token bucket burst	Up to b tokens

One-Line Definitions

- **Router:** Layer 3 device that routes packets between networks using IP addresses.
- **Gateway:** Protocol converter connecting networks with different protocols.
- **Subnetting:** Dividing a network into smaller subnets by borrowing host bits.
- **RIP:** Distance vector routing protocol; metric = hop count; max 15 hops.
- **OSPF:** Link state routing protocol; uses Dijkstra's algorithm; metric = cost.
- **BGP:** Path vector protocol for inter-AS routing on the Internet.
- **ARP:** Resolves IP address to MAC address via broadcast.
- **ICMP:** Reports network errors and provides diagnostic functions (ping).
- **TCP:** Reliable, connection-oriented transport protocol with flow/error control.

- **UDP:** Unreliable, connectionless transport protocol; low overhead, fast.
- **Congestion:** Network overload causing packet drops and retransmissions.
- **Leaky Bucket:** Traffic shaping algorithm enforcing constant output rate.
- **Token Bucket:** Traffic policing algorithm allowing controlled bursts.

Frequently Asked / Viva Questions

Q1. What is the difference between a switch and a router?

Switch operates at Layer 2 (MAC addresses, within a network); router operates at Layer 3 (IP addresses, between networks).

Q2. What is subnetting and why is it used?

Dividing a network into smaller subnets. Used to reduce broadcast domains, improve security, and efficient IP address utilisation.

Q3. How many hosts can a /26 subnet support?

$2^6 - 2 = 62$ usable hosts.

Q4. Difference between RIP and OSPF?

RIP: distance vector, hop count metric, max 15 hops, slow convergence. OSPF: link state, cost metric, no hop limit, fast convergence.

Q5. What is BGP used for?

Inter-domain routing between Autonomous Systems on the Internet; uses path vector algorithm.

Q6. What does ARP do?

Resolves an IP address to a MAC address by broadcasting an ARP request on the local network.

Q7. Difference between TCP and UDP?

TCP: reliable, connection-oriented, ordered, flow/error control. UDP: unreliable, connectionless, no ordering, minimal overhead.

Q8. What is the TCP three-way handshake?

SYN (client) → SYN-ACK (server) → ACK (client). Establishes a reliable connection.

Q9. What is the difference between leaky bucket and token bucket?

Leaky bucket: constant output rate, excess dropped. Token bucket: variable output up to burst size b , more flexible.

Q10. What is a choke packet?

A packet sent by a congested router directly to the source to reduce its transmission rate (closed-loop congestion control).

Frequently Asked / Viva Questions (contd.)**Q11. What is the purpose of TTL in IP header?**

Time To Live; decremented by each router. When $TTL = 0$, packet is discarded. Prevents infinite routing loops.

Q12. What are the main improvements of IPv6 over IPv4?

128-bit addressing, fixed 40-byte header, no router fragmentation, built-in IPSec, no broadcast (uses multicast).